DYLAN PIWKO

AEROSPACE ENGINEERING AT THE UNIVERSITY OF MICHIGAN

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ABOUT

I'm a first-year aerospace engineering student at the University of Michigan. I've been building and flying rockets for more than five years, and high-power rockets since 2023. These four projects are the ones I've learned the most from: a flight computer I'm designing, a NASA Ames concept study, a two-stage rocket that failed, and my first machine learning competition.

PROJECTS

Project	Role	Skills and tools	Page
Kestrel Flight Computer	Designer, Personal Project	KiCad, 4-layer PCB design, LoRa telemetry, GPS, verification planning	2
Aeropulse, NASA Ames Research Center	Concept Design Intern	Systems design, UAS route planning, UTM and airspace integration, technical writing	4
Terrier-Sandhawk Two-Stage Rocket	Builder and Flyer, Personal Project	RockSim, electronic staging, avionics integration, failure analysis	6
EUROCONTROL PRC Data Challenge	Team Member, Embry-Riddle Team	Python, LightGBM, CatBoost, SHAP, feature engineering	8

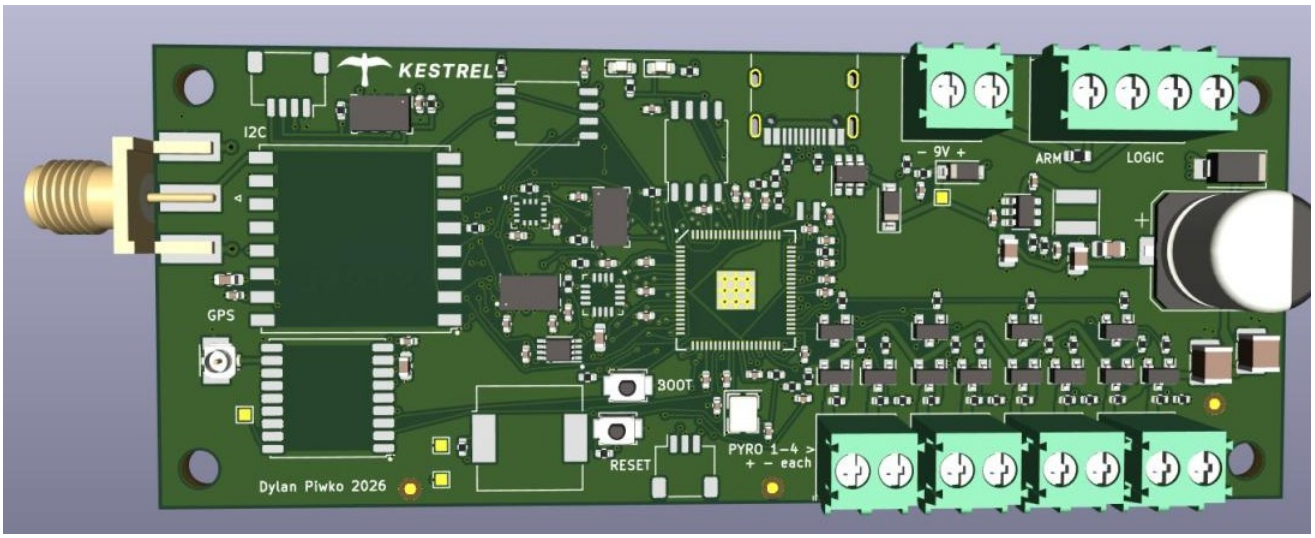


My Terrier-Sandhawk two-stage rocket at liftoff, January 2025

KESTREL FLIGHT COMPUTER

Designer, Personal Project
October 2026 – Present

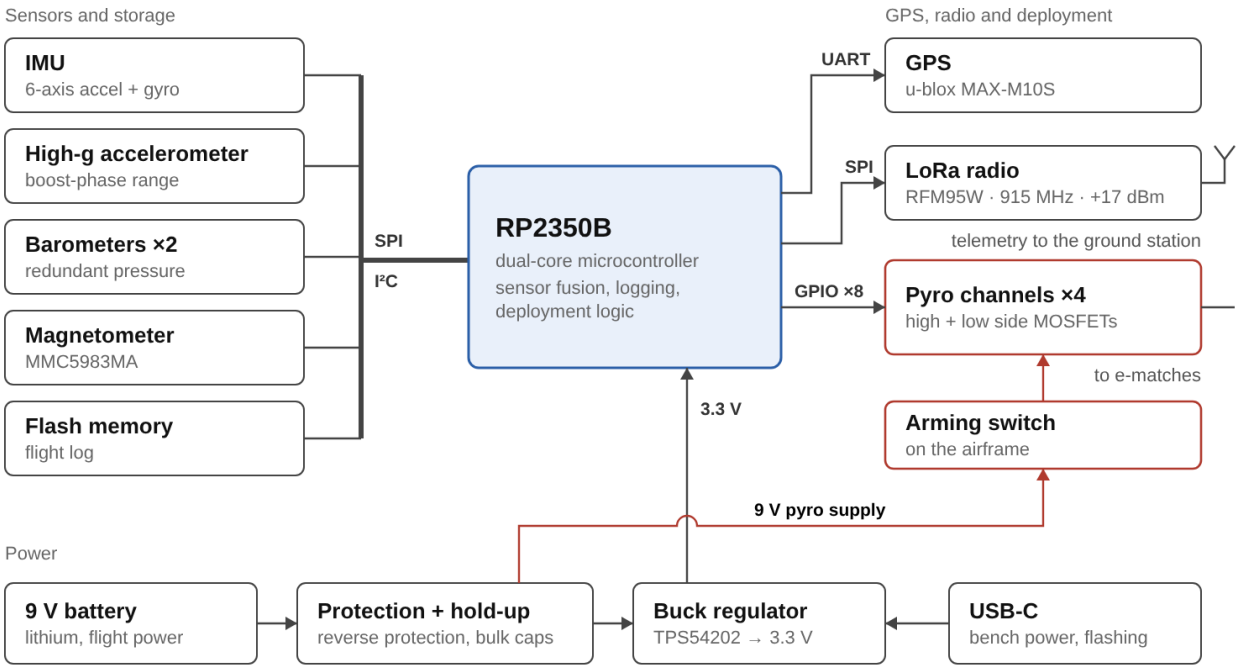
Designed and checked on paper. The first boards are being built now and haven't been tested. Kestrel is a 4-layer flight computer for my 4" Wildman Punisher 4. Its first flight, my NAR Level 2 attempt in January 2027, will be logging only.



Kestrel in KiCad: 100 × 40 mm, 4 layers, 145 parts. The terminals on the right are power, arming, and the four pyro channels.

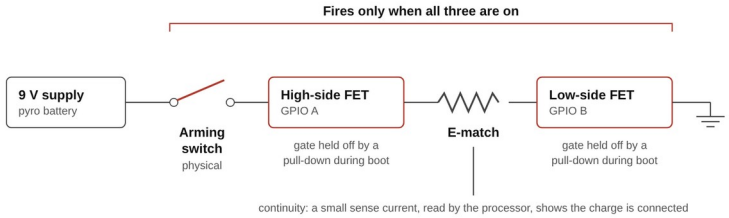
DESIGN DECISIONS

The real target is a later L2200G flight that simulates to about 14,800 ft and Mach 1.7. That flight should peak around 50 g, past the range of most IMU accelerometers, so I added a separate ±200 g accelerometer for boost. Two barometers give a cross-check on altitude, and GPS and a 915 MHz LoRa radio send position to the ground. Everything runs from a 9 V battery through a TPS54202 buck converter.



Block diagram. Every sensor, the radio, and the pyro channels connect to one RP2350B.

PYRO SAFETY



One pyro channel. Firing needs the arming switch and both transistors.

The pyro channels can hurt someone, so they got the most protection. Each e-match is switched on its high side and its low side, behind an arming switch on the airframe.

Pull-down resistors hold the gates off while the processor boots, and a small sense current checks continuity without firing.

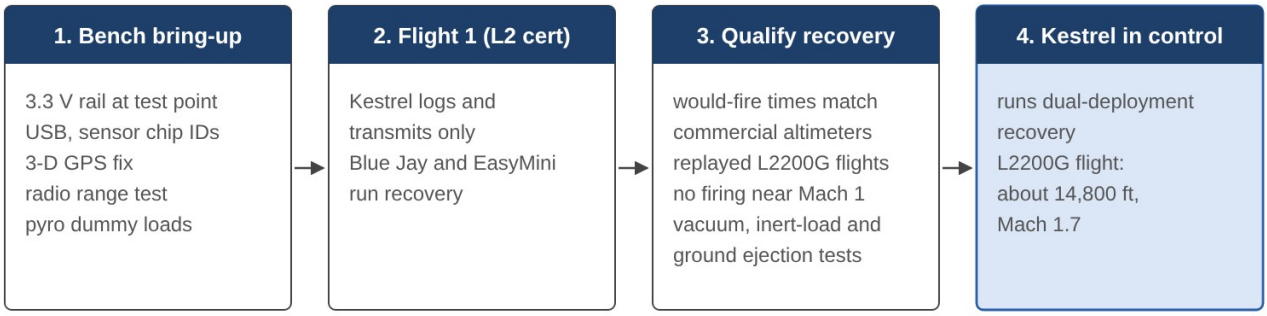
CHECKED BEFORE ORDERING

I could only afford one board revision, so I checked each subsystem against a datasheet or reference design and wrote down the test that will prove it. The design has zero ERC and DRC errors, PCBWay's engineers reviewed it, and PCBWay sponsored part of the order through their Future Innovators Program.

Subsystem	Checked against	Planned test once built
Power	TI's TPS54202 3.3 V reference values; USB-only start-up margin	3.25–3.35 V at the test point
Processor	All 80 RP2350B pins against the pin table and SparkFun's production board	Enumerates over USB
Sensors and GPS	Each pinout and SPI mode against its datasheet; u-blox integration manual	Chip ID self-test; 3-D GPS fix
Radio	Pinout; link budget of about 29 dB margin at 10 km	Ground-station range test
Pyro	MOSFET ratings, gate drive, boot pull-downs	Dummy-load tests, armed and disarmed
Layout	Zero ERC/DRC errors; PCBWay engineering review	Visual inspection of both boards

None of the tests in the right-hand column have been run yet.

BEFORE IT CONTROLS A PARACHUTE



The order Kestrel has to earn its way into controlling recovery

On flight 1, a Featherweight Blue Jay and an EasyMini handle recovery while Kestrel logs. Its would-fire times have to match when they actually fired. It also has to call deployments correctly on replayed L2200G simulations without firing through the Mach transition, then pass vacuum chamber, inert load, and ground ejection tests.

WHERE IT'S GOING

After the L2200G flight, I want to lower the cost per board enough to build them for more than one rocket, and maybe sell them. Boards with GPS, telemetry, and four or more pyro channels are expensive or come as two separate products:

Product	What it has	Price (Nov. 2026)
Altus Metrum TeleMetrum	2 pyro channels, GPS, telemetry	$300
Altus Metrum TeleMega	6 pyro channels, GPS, telemetry	$400
Featherweight Blue Raven	4 pyro channels, IMU, no GPS	$194.49
Featherweight GPS system	GPS tracker and ground station, no pyro channels	$420

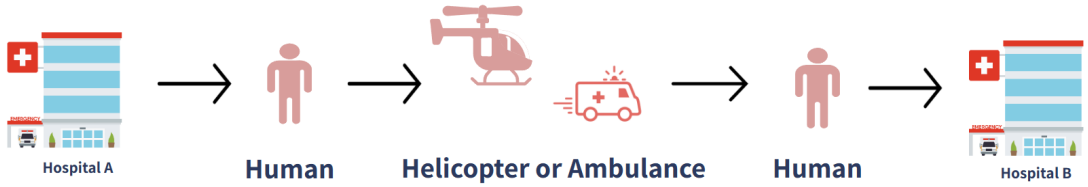
AEROPULSE, NASA AMES RESEARCH CENTER

Concept Design Intern, Aviation Systems Division
June – August 2025

In our model, a donor heart gets from San Francisco General to Santa Clara Valley Medical Center in about 37 minutes, compared with 70 to 90+ today. Our six-person team designed Aeropulse, a concept for moving organs and urgent medicine between hospitals with drones and stair-climbing robots. I was responsible for drone routing and urban airspace integration.

THE PROBLEM

A UNOS review of about 8,800 organ shipments from 2014 to 2019 found nearly 170 organs thrown out and 370 near misses because of transport delays of two hours or more. Most transfers go by road, and every handoff is another chance for a delay. Helicopters are used in fewer than 3% of cases because of cost.



Today: courier, vehicle, and hospital intake, with a handoff at each step (from the memo)



Aeropulse: a robot carries the payload to the roof, rides the drone, and delivers it inside Hospital B (from the memo)

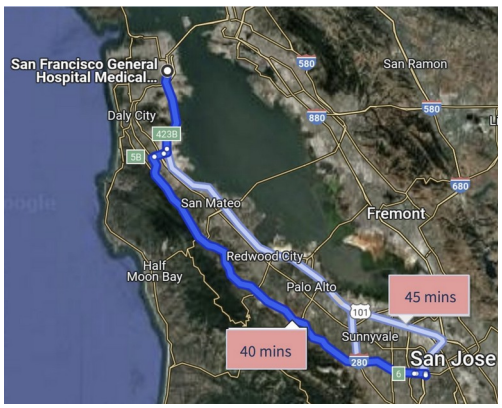
THE CONCEPT

A hybrid VTOL drone lifts the robot on a tethered hoist between hospital roofs. We based the drone on Sky-Drones' SkyLane, whose petrol cruise engine gives more range and wind tolerance than an all-electric drone.

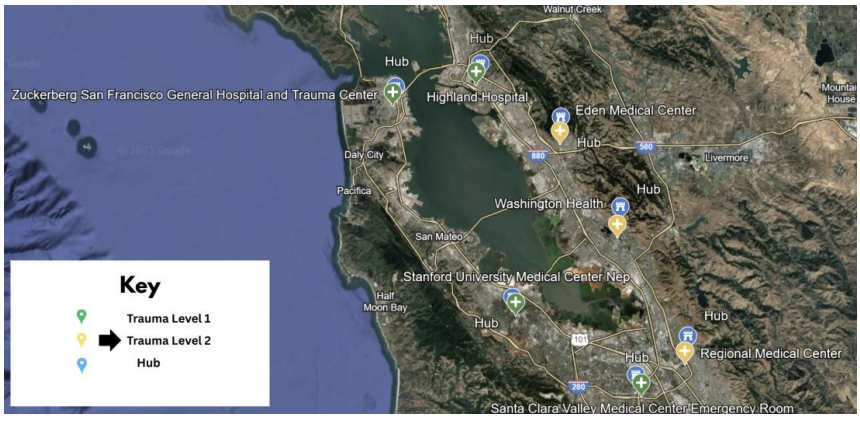
The pilot network we proposed had eight drones serving Bay Area trauma centers. Apart from a small 3D-printed model of the robot, nothing was built or flown.



The SkyLane hybrid VTOL we based the drone on (photo © Sky-Drones)



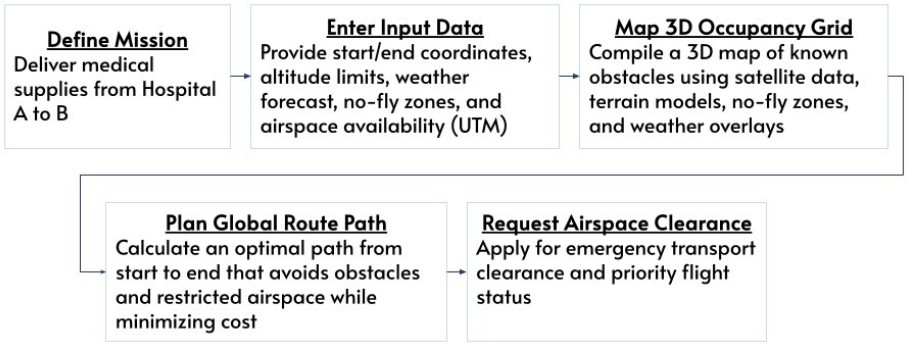
Modeled route, San Francisco General to Santa Clara Valley Medical Center (map data © Google)



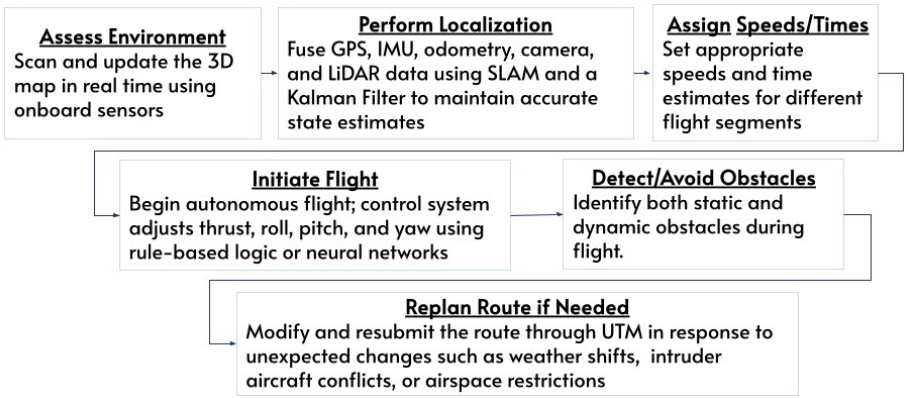
Proposed drone hubs near the trauma centers (imagery © Google)

MY PART: ROUTING THROUGH CITY AIRSPACE

Before takeoff, the drone plans its whole route. It builds a 3-D occupancy grid from altitude limits, weather, no-fly zones, and open UTM airspace, finds the lowest-cost path, and requests emergency priority clearance through UTM.



Pre-flight planning ends with the UTM clearance request (from the memo).

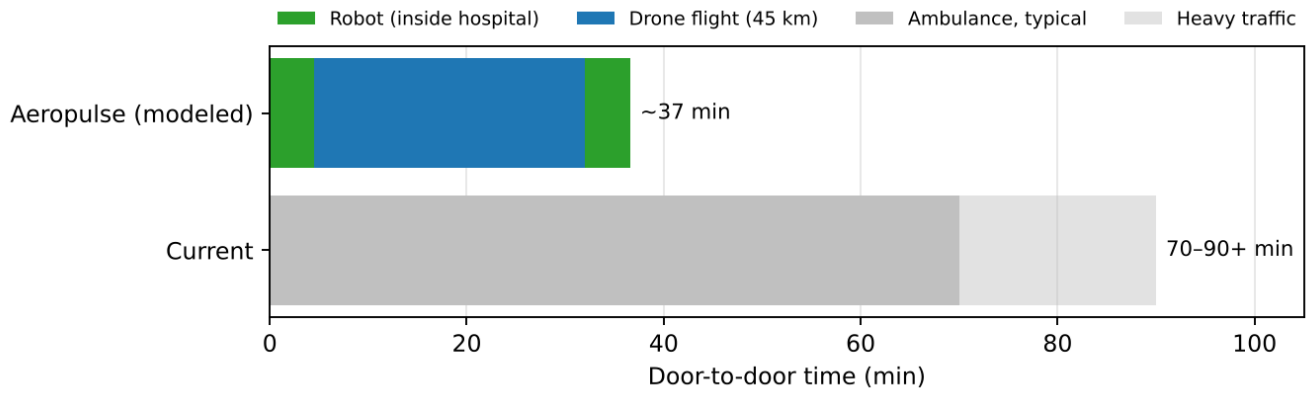


The in-flight loop, including detect-and-avoid and replanning (from the memo).

In the air, it localizes with GPS, IMU, odometry, camera, and LiDAR through SLAM and a Kalman filter, and replans through UTM when weather or traffic changes.

If it can't continue safely, it can loiter, land somewhere low-risk, return to base, or deploy a parachute. We planned for flights below 400 ft AGL under Part 107 with BVLOS waivers.

OUTCOME



Modeled heart transfer, door to door. Both bars are estimates, not measured trips.

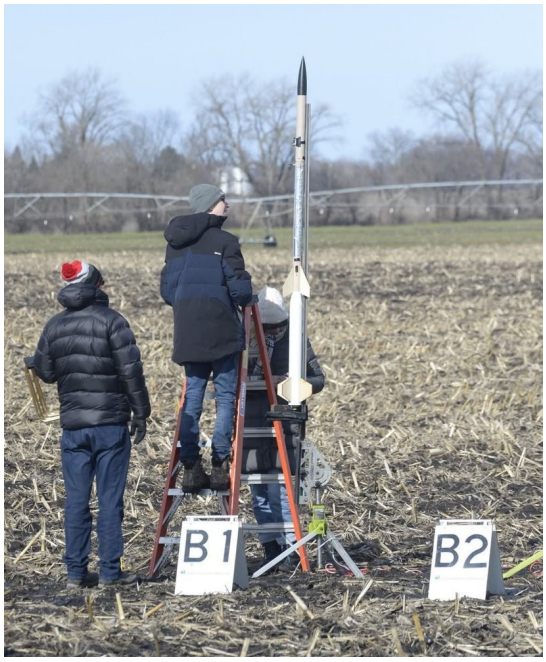
For the timing model we used 110 km/h for the drone, below its quoted 80-knot cruise, to leave room for airspace restrictions and coordination. It's a first-order estimate. We presented the design to more than 50 NASA staff and published it as NASA/TM-20250009029, with all six of us listed as equal authors.

Full memo: ntrs.nasa.gov/citations/20250009029

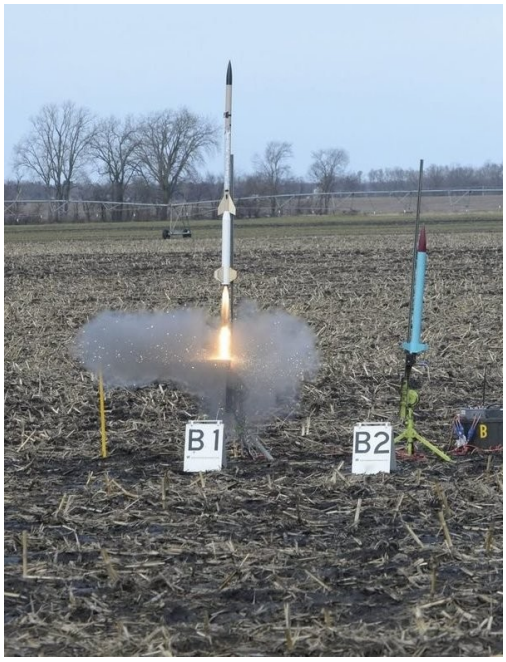
TERRIER-SANDHAWK TWO-STAGE ROCKET

Builder and Flyer, Personal Project
January 2025

The stages didn't separate, the sustainer never lit, and the upper stage was lost. The most likely cause is one loose battery mount shared by both altimeters. This was my first two-stage rocket: a LOC Precision 2.26" Sandhawk sustainer on a Terrier booster, flown at Michiana Rocketry on a CTI I297 staging to an H180.



Setting up on pad B1

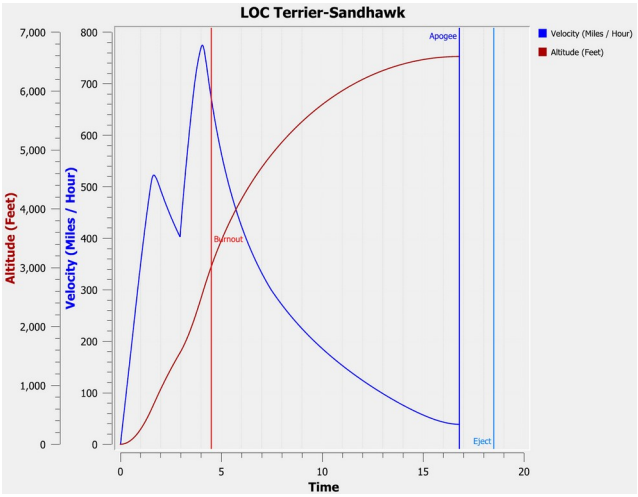


Liftoff, January 31, 2025

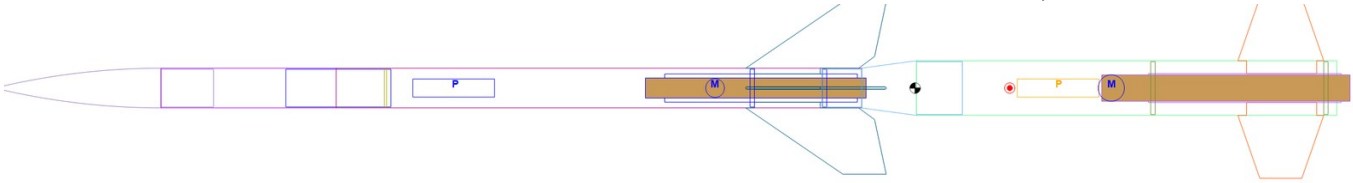
DESIGN AND STAGING

RockSim gave a 2.40 static margin and about 6,590 ft. In the sustainer, an Altus Metrum EasyTimer ran staging and an EasyMini ran recovery, with a single main parachute held closed by a Jolly Logic Chute Release until 400 ft. The igniter leads ran down the outside of the airframe under aluminum tape.

I picked the staging delays myself: 1.0 s from booster burnout to separation, then 0.7 s to sustainer ignition. Longer gaps make each event easier to see from the ground and can add altitude, but the sustainer lights slower and is easier for wind to push off course. I wanted it to still fly well in imperfect weather.



RockSim prediction: about 6,590 ft and 775 mph, with the sustainer lit at separation



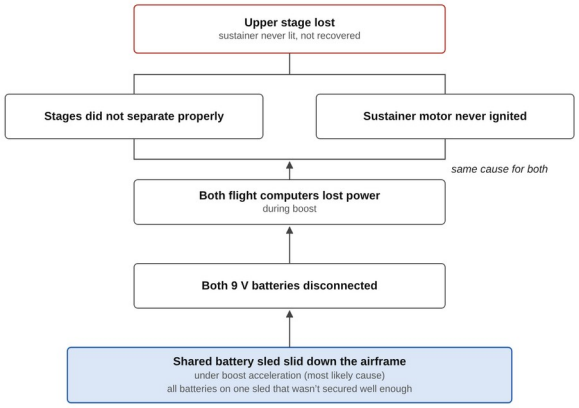
RockSim model, sustainer on the left. CG at 52.5 in and CP at 57.9 in from the nose.

FAILURE ANALYSIS

Both failures happened at once, so I looked for one cause. Both altimeters ran on 9 V batteries mounted to one balsa sled, inside a coupler held by two plastic rivets.

Either the sled broke or the rivets sheared. Under boost the batteries would slide aft and pull loose from both altimeters at the same moment.

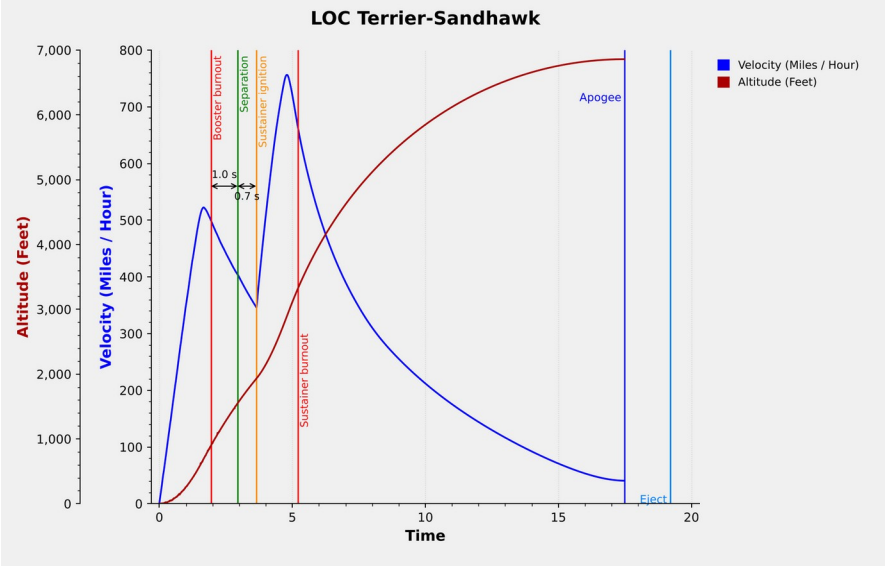
I can't confirm it because the upper stage was never found. It taught me that two flight computers aren't redundant if they share one battery mount.



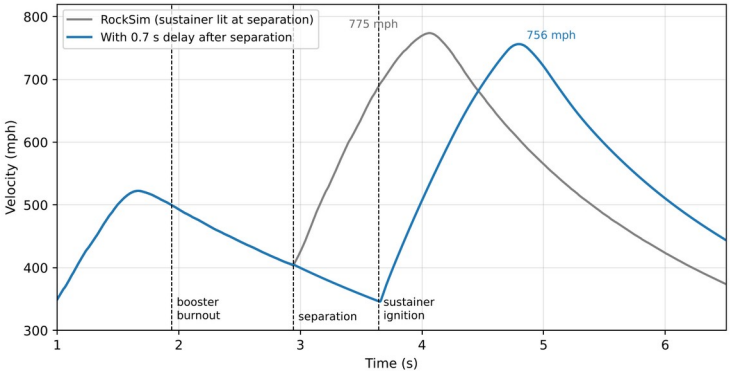
Fault tree from the lost upper stage back to the shared battery sled

CHECKING THE STAGING PREDICTION

RockSim lights the sustainer at separation, so it ignores the 0.7 s delay. The upper stage was re-flown from separation with a 2-D point-mass model, using RockSim's drag and the H180's thrust curve. With RockSim's own timing, the model matches its apogee within 3 ft.



With both delays, booster burnout, separation, and sustainer ignition are spaced out, and apogee rises to about 6,860 ft.



Sustainer velocity: RockSim (gray) vs. with the 0.7 s delay (blue)

Lighting at 347 mph instead of 405 mph keeps more of the burn below Mach 1, where drag is highest.

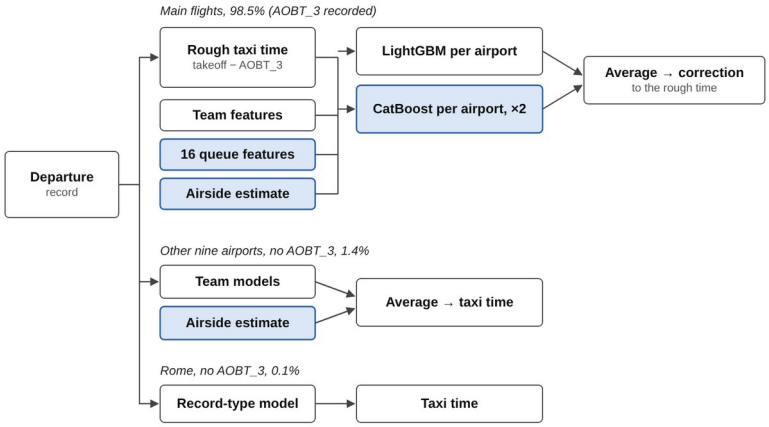
A 10% change in drag moves the prediction by about 300 ft. With no sustainer flight, there's no data to check it against.

EUROCONTROL PRC DATA CHALLENGE

Team Member, Embry-Riddle Team
September – October 2026

My model lowered the team's official RMSE from 276.53 s to 267.34 s, and SHAP showed that most of our worst predictions came from bad timestamps. The challenge was to predict taxi-out time at ten European airports from about 2 million 2025 departures. I joined with very little Python or machine learning experience and was responsible for the main-flight model, 98.5% of departures.

The team's model has separate paths for main flights, the other airports' flights without a reported off-block time, and Rome. My additions, in blue, are all in the main-flight path.

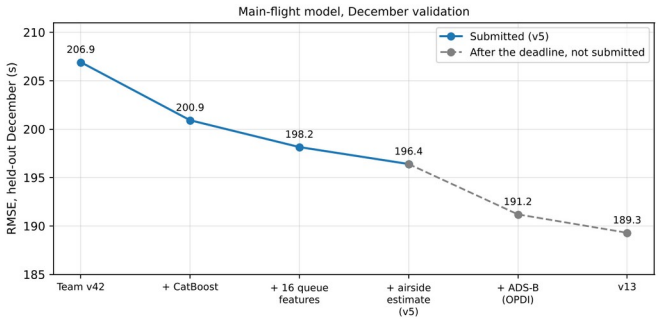


Team model, my additions in blue

TESTING ONE CHANGE AT A TIME

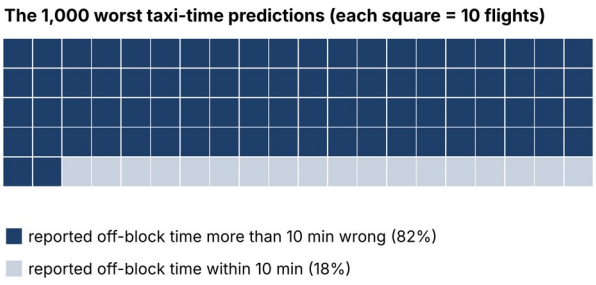
Each change was tested on held-out December data and had to gain at least 0.5 s and help at 6 of 10 airports. Three made it into my submitted model:

- CatBoost alongside the team's LightGBM, from Dr. Garcia's feedback: 206.9 to 200.9 s
- 16 departure-queue features: 200.9 to 198.2 s
- An airside taxi-time estimate: 198.2 to 196.4 s



Held-out December error by version. Gray points came after the deadline.

FINDING THE BAD TIMESTAMPS



Most of the worst 1,000 predictions had a bad off-block time.

Dr. Li suggested SHAP. Of the 1,000 worst predictions, 82% had a reported off-block time more than 10 minutes wrong, against 6% of all flights.

So I trained an airside estimate that doesn't use that timestamp and gave it to the model as a second opinion.

RESULTS

Taeyun Yoo built the team's best leaderboard score, 265.02 s, and the final pipeline on top of my model, which still uses my CatBoost models and airside estimate. ADS-B features I tried after the deadline helped further but weren't submitted. We finished 47th of 94 at 220.23 s, on a February and June final set that isn't directly comparable to the leaderboard. The top teams used raw ADS-B tracks, which address the same timestamp problem.

Team code: github.com/taeyunyoo/prc-2026-merry-llama